RELATIVE R-CYCLES DESCEND THROUGH LIMITS OF NOETHERIAN SCHEMES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The locus where two relative cycles on the fibres agree is closed (relative-cycles chapter, Lemma lemma-relative-cycles-equal, canonical tag 0H5C). cited at proof lines 1398-1400. Two cycles mapping to the same element at the limit base change into its closed locus E in S_i.
- Descent of constructible containment along limits: if the limit morphism maps into a constructible set then some transition morphism does (Limits chapter, Lemma limits-lemma-limit-contained-in-constructible, canonical tag 05F4). cited at proof lines 1401-1403. Provides the index j at which the transition morphism lands in E, proving injectivity.
- Proper completely decomposed covers putting a relative cycle into the image of the cycle-classes map (relative-cycles chapter, Lemma lemma-get-cycles, canonical tag 0H59) and the cycle-classes construction itself (relative-cycles chapter, Equation equation-cycle-classes, canonical tag 0H58). cited at proof lines 1408-1413. Reduces surjectivity to cycles of the form sum n_a [Z_a/X'/S']_r with Z_a flat of relative dimension at most r.
- Descending relative algebra through limits: descent of finite presentation (limits-lemma-descend-finite-presentation, 01ZM), of closed immersions of finite presentation (limits-lemma-descend-closed-immersion-finite-presentation, 01ZP), of flatness of finite presentation (limits-lemma-descend-flat-finite-presentation, 04AI), eventual properness (limits-lemma-eventually-proper, 081F), and dimension of the limit (limits-lemma-limit-dimension, 05M5), gathered in Limits, Section limits-section-descending-relative (canonical tag 01ZL). cited at proof lines 1416-1434. Produce an index i with a completely decomposed proper morphism g_i : S'_i to S_i descending g and closed subschemes Z_{ai} flat of relative dimension at most r descending the Z_a.
- Descent of completely decomposed morphisms (More on Morphisms, Lemma more-morphisms-lemma-descend-cd, canonical tag 0GTM). cited at proof lines 1433-1434. Makes the descended g_i completely decomposed.

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- Descent of families of r-cycles along morphisms with separable residue field extensions (relative-cycles chapter, Lemma lemma-descend-family, canonical tag 0H4X). cited at proof lines 1451-1455 and 1458-1459. Its existence and uniqueness produce the unique family alpha_i with g_i^*alpha_i = alpha'_i and force the image of alpha_i at the limit to be alpha.
- H-descent for the relative cycle property (relative-cycles chapter, Lemma lemma-relative-cycles-h-descent, canonical tag 0H52). cited at proof lines 1456-1457. Promotes the descended family alpha_i to an element of z(X_i/S_i, r).

Source reference. Original lemma 0H5C: sources/stacks-chow/tags.txt, source lines 21382–21382. Cited at proof lines 1398-1400: the closed locus E into which the limit morphism maps in the injectivity paragraph.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cited at proof lines 1401-1403: produces the index j with the transition morphism inside E.

Source reference. Original lemma 0H59: sources/stacks-chow/tags.txt, source lines 21379–21379. Cited at proof lines 1408-1410: starts the surjectivity paragraph.

Source reference. Original equation 0H58: sources/stacks-chow/tags.txt, source lines 21378–21378. Cited at proof lines 1410 and 1411: the image of which g^*alpha is an element.

Source reference. Original section 01ZL: sources/stacks-chow/tags.txt, source lines 2598–2598. Cited at proof lines 1416-1417: the machinery used to descend the cover and the flat subschemes.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cited at proof lines 1427-1428: descends the morphisms of finite presentation in the diagram.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cited at proof lines 1429-1430: descends the closed immersions cutting out the Z_a.

Source reference. Original lemma 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cited at proof lines 1430-1431: keeps the descended Z_{ai} flat of relative dimension at most r.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cited at proof lines 1431-1432: descends the properness of the cover g.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Cited at proof lines 1432-1433: controls the relative dimension bound of the descended subschemes.

Source reference. Original lemma 0GTM: sources/stacks-chow/tags.txt, source lines 20972–20972. Cited at proof lines 1433-1434: makes the descended g_i completely decomposed.

Source reference. Original lemma 0H4X: sources/stacks-chow/tags.txt, source lines 21367–21367. Cited at proof lines 1451-1455 and 1458-1459: existence and uniqueness of the descended family alpha_i.

Source reference. Original lemma 0H52: sources/stacks-chow/tags.txt, source lines 21372–21372. Cited at proof lines 1456-1457: alpha_i lies in z(X_i/S_i, r).

Lemma 0.2. *Let $S = \lim_{i \in I} S_i$ be the limit of a directed inverse system of Noetherian schemes with affine transition morphisms. Let $0 \in I$ and let $X_0 \to S_0$ be a finite type morphism of schemes. For $i \geq 0$ set $X_i = S_i \times_{S_0} X_0$ and set $X = S \times_{S_0} X_0$. If S is Noetherian too, then*

$$z(X/S, r) = \operatorname{colim}_{i \geq 0} z(X_i/S_i, r)$$

where the transition maps are given by base change of relative r-cycles.

Proof. Suppose that $i \geq 0$ and $\alpha_i, \beta_i \in z(X_i/S_i, r)$ map to the same element of $z(X/S, r)$. Then $S \to S_i$ maps into the closed subset $E \subset S_i$ of Lemma 0H5C. Hence for some $j \geq i$ the morphism $S_j \to S_i$ maps into E, see Limits, Lemma 05F4. It follows that the base change of α_i and β_i to S_j agree. Thus the map is injective.

Let $\alpha \in z(X/S, r)$. Applying Lemma 0H59 a completely decomposed proper morphism $g : S' \to S$ such that $g^*\alpha$ is in the image of (0H58). Set $X' = S' \times_S X$. We write $g^*\alpha = \sum n_a [Z_a/X'/S']_r$ for some $Z_a \subset X'$ closed subscheme flat and of relative dimension $\leq r$ over S'.

Now we bring the machinery of Limits, Section 01ZL ff to bear. We can find an $i \geq 0$ such that there exist

(1) a completely decomposed proper morphism $g_i : S_i' \to S_i$ whose base change to S is $g : S' \to S$,
(2) setting $X_i' = S_i' \times_{S_i} X_i$ closed subschemes $Z_{ai} \subset X_i'$ flat and of relative dimension $\leq r$ over S_i' whose base change to S' is Z_a.

To do this one uses Limits, Lemmas 01ZM, 01ZP, 04AI, 081F, and 05M5 and More on Morphisms, Lemma 0GTM. Consider $\alpha_i' = \sum n_a [Z_{ai}/X_i'/S_i']_r \in z(X_i'/S_i', r)$. The image of α_i' in $z(X'/S', r)$ agrees with the base change $g^*\alpha$ by construction.

Set $S_i'' = S_i' \times_{S_i} S_i'$ and $X_i'' = S_i'' \times_{S_i} X_i$ and set $S'' = S' \times_S S'$ and $X'' = S'' \times_S X$. We denote $\mathrm{pr}_1, \mathrm{pr}_2 : S'' \to S'$ and $\mathrm{pr}_1, \mathrm{pr}_2 : S_i'' \to S_i'$ the projections. The two base changes $\mathrm{pr}_1^*\alpha_i'$ and $\mathrm{pr}_1^*\alpha_i'$ map to the same element of $z(X''/S'', r)$ because $\mathrm{pr}_1^* g^*\alpha = \mathrm{pr}_1^* g^*\alpha$. Hence after increasing i we may assume that $\mathrm{pr}_1^*\alpha_i' = \mathrm{pr}_1^*\alpha_i'$ by the first paragraph of the proof. By Lemma 0H4X we obtain a unique family α_i of r-cycles on fibres of X_i/S_i with $g_i^*\alpha_i = \alpha_i'$ (this uses that $S_i' \to S_i$ is completely decomposed). By Lemma 0H52 we see that $\alpha_i \in z(X_i/S_i, r)$. The uniqueness in Lemma 0H4X implies that the image of α_i in $z(X/S, r)$ is α and the proof is complete. □